

Identifying the triangular-lattice compound YbZnGaO_4 as a spin glass

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YbMgGaO_4 with the triangular-lattice structure has been reported to be a promising quantum-spin-liquid candidate. However, here we provide compelling evidence that its sister compound YbZnGaO_4 is a spin glass. We show that the spin-glass phase can give rise to all of our experimental observations, including no long-range magnetic order, prominent broad excitation continua, and absence of magnetic thermal conductivity. More crucially, from the ultralow-temperature a.c. susceptibility measurements, we unambiguously observe frequency-dependent peaks around 0.1 K, indicating the spin-glass ground state. We consider disorder and frustration to be the main driving force for this phase.

Quantum spin liquids (QSLs) represent a novel state of matter in which spins are highly entangled, but neither order nor freeze at low temperatures[1, 2]. There is accumulating experimental evidence suggesting that such a state is realized in YbMgGaO_4 (refs 3–11). The magnetic specific heat C_m is proportional to T^α with $\alpha \approx 2/3$ (refs 3, 5, and 12). It has a negative Curie-Weiss temperature of $\Theta \sim -4$ K (refs 3 and 4) but does not show a long-range magnetic order at low temperatures[5, 6]. Moreover, diffusive continuous magnetic excitations have been observed by inelastic neutron scattering (INS) measurements[5, 6], which are interpreted as resulting from the fractional spin excitations of a QSL (refs 13 and 14). However, there are also reports challenging this idea: i) The thermal conductivity (κ) study in ref. 12 reveals no contributions to κ from magnetic excitations despite the large magnetic specific heat at low temperatures, casting doubts on the existence of itinerant quasiparticles expected for a QSL (ref. 15); ii) Since Mg^{2+} and Ga^{3+} in the nonmagnetic layers are randomly distributed[3, 4, 16], the disorder effect, which is detrimental to the QSL phase for this compound[17], can be significant[5, 8].

In this Letter, we report comprehensive measurements on a closely related system, YbZnGaO_4 . We show that the most natural conclusion, that is consistent with the micro- and macro-scopic data presented here is that the system is a spin glass. We believe disorder[5, 8, 11, 16, 17] and frustration[4, 5, 11, 18–22] to be largely responsible for this phase.

High-quality single crystals of YbZnGaO_4 were grown by the floating-zone technique, overcoming the problem caused by the volatile nature of ZnO (refs 3 and 23).

The d.c. and a.c. magnetic susceptibility and specific heat were measured in a Quantum Design physical property measurement system. INS experiments on the single crystals were carried out on PANDA located at MLZ at Garching, Germany[24]. In the measurements, the 11 co-aligned single crystals weighed 1.2 g in total with a sample mosaic of 0.98° were mounted in the $(H, K, 0)$ plane. INS experiments on the 14-g powder sample were carried out on SIKA located at ANSTO at Lucas Heights, Australia. The wave vector $\mathbf{Q}$ was expressed as (H, K, L) reciprocal lattice unit (r.l.u.) of $(a^*, b^*, c^*) = (4\pi/\sqrt{3}a, 4\pi/\sqrt{3}b, 2\pi/c)$, with $a = 3.414(2)$ Å and $c = 25.140(2)$ Å.

YbZnGaO_4 is isostructural to YbMgGaO_4 , both of which have the YbFe_2O_4 -type structure (space group $R\bar{3}m$, No. 166) (refs 23, 25, and 26). Schematics of the crystal structure and two-dimensional triangular lattice of Yb^{3+} are illustrated in Fig. 1(a) and (b), respectively. The magnetic ground state of Yb^{3+} ions is an effective spin-1/2 Kramers doublet[3, 5, 8]. In YbZnGaO_4 , the d.c. magnetic susceptibility of the effective spin follows the Curie-Weiss law from 2 to 30 K. In the inset of Fig. 1(c), we show the inverse susceptibility and the Curie-Weiss fits up to 30 K. From the fits, we find Θ to be $-2.70(2)$, $-2.38(3)$, and $-2.46(2)$ K, for the single crystal with magnetic fields perpendicular and parallel to the c axis and for the polycrystalline sample, respectively. The negative sign shows that the magnetic ground state is dominated by antiferromagnetic interactions. The superexchange coupling constant J is estimated to be $1.73(5)$ K. These parameters are summarized in Table I, together with other values for this material, in comparison with YbMgGaO_4 .

TABLE I. Some parameters for YbZnGaO₄ and YbMgGaO₄.

Compound	$\Theta_{\perp}$ (K)	$\Theta_{\parallel}$ (K)	Θ_p (K)	J (K)	$g_{\perp}$	$g_{\parallel}$	g_p	T^+ (K)	α	Δ (K)	T_f (K)	ΔP
YbZnGaO ₄	-2.70(2)	-2.38(3)	-2.46(2)	1.73(5)	3.17(4)	3.82(2)	3.58(3)	1.86(5)	0.59(2)	6.18(3)	0.093(6)	0.053(2)
YbMgGaO ₄	-4.78	-3.20	-4.11	1.5	3.00	3.82	3.21	2.40	0.74	8.26	0.099(6)	0.068(4)

$\Theta_{\perp}$, $\Theta_{\parallel}$, and Θ_p are Curie-Weiss temperatures for the single crystal with magnetic fields perpendicular and parallel to the c axis, and for the polycrystalline sample, respectively. J is the superexchange coupling constant, approximated by[4] $J = (4J_{\perp} + J_{zz})/3$, where $J_{\perp} \approx -\Theta_{\perp}/3 = 0.90(1)$ K, and $J_{zz} \approx -2\Theta_{\parallel}/3 = 1.59(2)$ K. $g_{\perp}$, $g_{\parallel}$, and g_p are Landé g factors, obtained by fitting the magnetization data in Fig. S2. T^+ is the peak temperature of the zero-field magnetic specific heat. α is the fitted index using $C_m \sim T^{\alpha}$. Δ is the magnon gap obtained by fitting the 9-T data with $C_m \sim \exp(-\Delta/T)$. Corresponding values for YbMgGaO₄ are from refs 3, 4, 6, and 12.

T_f is the peak temperature of the real part of the a.c. susceptibility (χ') at 100 Hz. ΔP is the peak shift $\frac{\Delta T_f}{T_f \Delta \lg(f)}$ from 100 to 10000 Hz. These two parameters for both YbZnGaO₄ and YbMgGaO₄ are obtained from our own measurements.

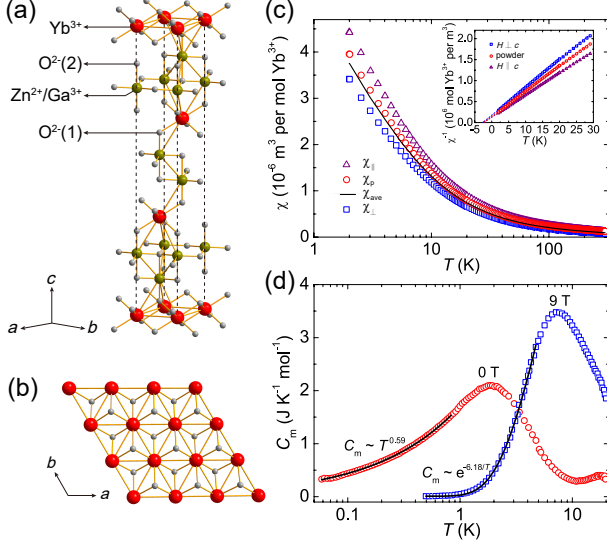


FIG. 1. (a) Schematic crystal structure for YbZnGaO₄ and YbMgGaO₄. (b) Top view of the triangular layer of YbO₆ octahedra. (c) D.C. magnetic susceptibility for fields applied parallel ($\chi_{\parallel}$) and perpendicular ($\chi_{\perp}$) to the c axis for the single crystal, and polycrystalline sample (χ_p), measured with a 0.1-T field. The data have been corrected by the Van Vleck paramagnetic susceptibility, as discussed in the Supplementary Materials. The solid line is the averaged susceptibility χ_{ave} . The inset shows the inverse susceptibility and their accompanying Curie-Weiss fits. (d) Magnetic specific heat (C_m) measured at zero and 9-T fields. C_m is obtained by subtracting the contribution from the lattice using a nonmagnetic reference sample LuZnGaO₄. Solid lines are fits to the data described in the main text.

In Fig. 1(d) we plot the magnetic specific heat (C_m) down to 0.05 K for YbZnGaO₄. From the zero-field data, we do not observe a λ -type peak expected for a well-defined phase transition. Instead, there is a broad peak at $T^+ \approx 1.86(5)$ K, below which C_m decreases. Below T^+ , we fit C_m to T^{α} and determine α to be 0.59(2). We have also attempted to fit the low-temperature data using $C_m \sim \exp(-\Delta/T)$, and obtained a small gap of 0.05(3) K, consistent with the large magnetic specific

heat arising from the gapless magnetic excitations at low temperatures. With increasing fields, T^+ gets higher, and the hump becomes narrower. At 9 T, T^+ should correspond to a transition from the paramagnetic to ferromagnetic state, as the system is in a fully polarized state at low temperatures [see Fig. S2(a)]. When we fit the 9-T data with $C_m \sim \exp(-\Delta/T)$, we obtain $\Delta = 6.18(3)$ K, which corresponds to a magnon gap induced by an external magnetic field, as also observed in YbMgGaO₄ (refs 5 and 12).

We now explore the system by carrying out INS experiments, which reveal similar behaviors to YbMgGaO₄ (refs 5 and 6). In Fig. 2(a) and (b), we present the contour maps of the excitation spectra at energy transfers of $E = 0.3$ and 0.6 meV, respectively. The broad diffusive excitations spreading along edges of the two-dimensional (2D) Brillouin zone, and decreasing in intensity with increasing Q , indicate that the system is dominated by antiferromagnetic correlations but without long-range order, consistent with the macroscopic results in Fig. 1(c) and (d). Magnetic dispersions along two high-symmetry paths are plotted in Fig. 2(c), which exhibit a continuum over the whole energy range measured. The excitations are gapless (see also, Fig. S4), consistent with the specific heat data. Constant-energy scans along M_1 -K- Γ_1 at two representative energies are shown in Fig. 2(d); similar scans along Γ_1 -M₂- Γ_2 are shown in Fig. 2(e). At $E = 0.8$ meV, intensities remain roughly constant from M_1 to $(1/4, 1/2, 0)$, and then decrease as Q approaches Γ_1 . The scan along the Γ_1 -M₂- Γ_2 direction results in a broad peak centering at M_2 , and the spin-spin correlation length is estimated to be 3 Å from this scan. This length scale is close to those obtained in YbMgGaO₄ (refs 5 and 6) and other QSL candidates[13, 14].

As is shown in refs 5 and 6, a QSL phase can give rise to the observed INS spectra. However, we notice that the cations in the nonmagnetic layers are randomly distributed[3, 4, 8, 11, 16]. As a result, there should be a strong variation in the magnetic couplings due to the disordered charge environment[8, 17]. In addition, the small J will further exaggerate the disorder effect. Can disorder

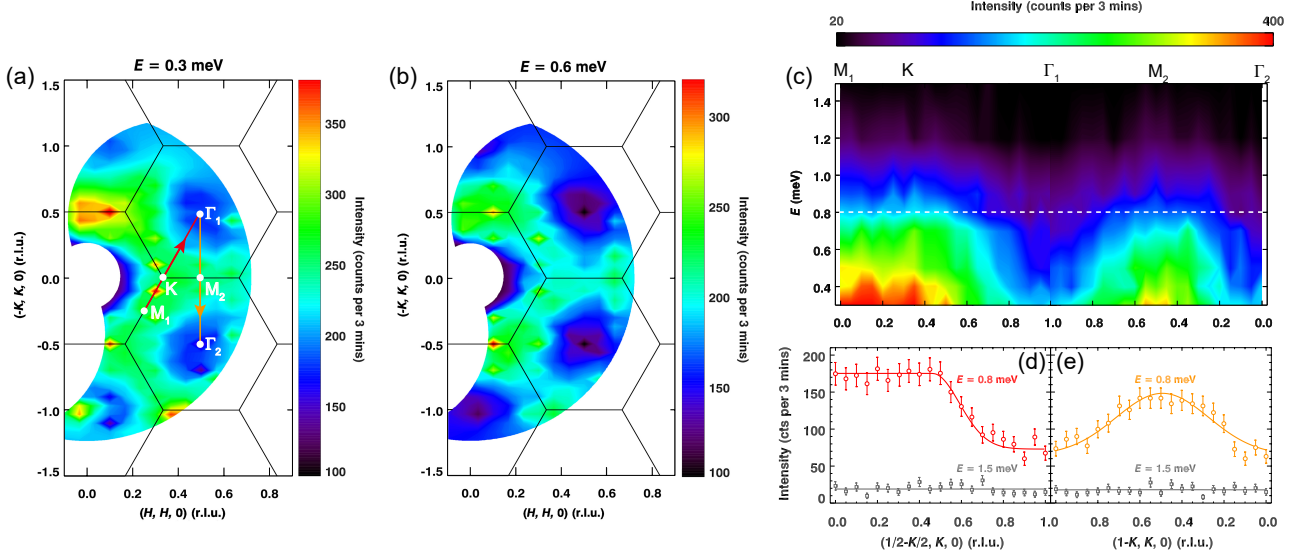


FIG. 2. (a) and (b) are contour maps of the INS spectra at $E = 0.3$ and 0.6 meV, respectively, measured at $T = 0.47$ K. The maps are obtained by plotting together a series of constant-energy scans along the $[H, 0, 0]$ direction with a step size of 0.1 r.l.u., and an interval of 0.1 r.l.u. along the $[0, K, 0]$ direction. Solid lines indicate Brillouin zone boundaries. The additional bright feature around $(-0.5, 0.5, 0)$ in (a) is due to the elastic scattering of the $(-1, 1, 0)$ nuclear Bragg peak (see Supplementary Materials for details). (c) Magnetic dispersion along M_1 -K- Γ_1 and Γ_1 - M_2 - Γ_2 directions as illustrated by the arrows in (a). The dispersion is obtained by plotting together a series of constant-energy scans as shown in (d) and (e), with an energy interval of 0.1 meV. The dashed line indicates constant-energy scans at $E = 0.8$ meV. In (d) and (e), lines through data are guides to the eye. Errors represent one standard deviation throughout the paper.

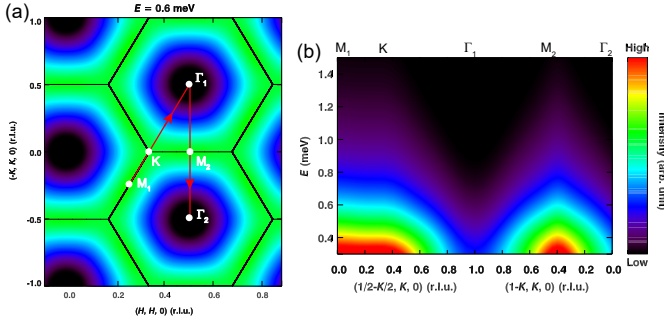


FIG. 3. (a) Contour map of the calculated spectra at $E = 0.6$ meV. (b) Calculated dispersions along the two high-symmetry paths illustrated in (a).

der make the magnetic excitations mimic those expected for a QSL (ref. 17)? In this context, we consider introducing disorder into a stripe-order phase, which is suggested to be the ground state for YbMgGaO_4 in the absence of disorder[17, 27]. We use an anisotropic spin model with nearest-neighbor and next-nearest-neighbor exchange interactions, and perform calculations with the linear spin-wave theory[28]. Without disorder, gapless spin-wave excitations disperse up from the M point. With increasing disorder, the well-defined spin-wave dispersions become broader both in momentum and energy. An example is presented in Fig. 3. The calculated results agree with

the experimental data quite well, demonstrating that an antiferromagnet with disorder can also exhibit the continuum-like INS spectra.

We further show thermal conductivity (κ) results in Fig. 4(a). At $T = 0.1$ K, κ is only about half of that of the nonmagnetic sample, LuMgGaO_4 , in which only phonons contribute to κ . This reduction is quite likely due to the scattering of phonons off the gapless magnetic excitations[12]. This also manifests itself in the magnetic-field measurements: in a field of 9 T that opens a gap of $6.18(3)$ K, there are almost no magnetic excitations to scatter phonons, so κ increases. We fit the zero-field data with $\kappa/T = \kappa_0/T + nT^{\beta-1}$ up to 0.4 K. Here, the first term κ_0 and second term nT^{β} represent non-phonon and phonon contributions, respectively. From the fit, we obtain $\kappa_0/T = -0.011(2)$ mW K $^{-2}$ cm $^{-1}$, and $\beta = 1.97(2)$. In the nonmagnetic sample LuMgGaO_4 , it is shown that $\kappa_0/T = -0.007$ mW K $^{-2}$ cm $^{-1}$, and $\beta = 2.07$ (ref. 12). In both materials, κ_0/T is virtually zero, similar to the case of YbMgGaO_4 (ref. 12). In contrast, another QSL candidate $\text{EtMe}_3\text{Sb}[\text{Pd}(\text{dmit})_2]_2$ has a high $\kappa_0/T = 2$ mW K $^{-2}$ cm $^{-1}$, considered to be a signature of highly mobile quasiparticles in the QSL state[15]. We therefore believe that a gapless QSL is not an applicable description for YbZnGaO_4 , because its magnetic excitations will contribute to κ (refs 29 and 30). On the other hand, the thermal conductivity results can be understood within a disordered-magnet picture,

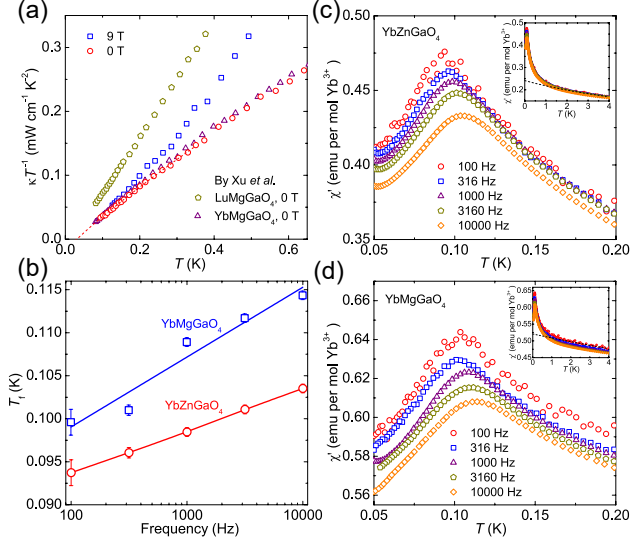


FIG. 4. (a) Thermal conductivity results on YbZnGaO₄ under zero and 9-T magnetic fields applied parallel to the *c* axis. The dashed line is a fit to the data described in the main text. For comparison, results on YbMgGaO₄ and the nonmagnetic reference compound LuMgGaO₄ are also plotted[12]. (b) Frequency dependence of the freezing temperature for both YbZnGaO₄ and YbMgGaO₄, extracted from the temperature dependence of the real part of the a.c. susceptibility (χ') shown in (c) and (d). Lines through data are guides to the eye. In the insets of (c) and (d), χ' in an extended temperature range up to 4 K are plotted. Dashed lines indicate the Curie-Weiss fits for the 100-Hz data.

in which the mean-free path of the magnons is reduced with disorder, and they are not expected to conduct heat.

Taking all aforementioned observations into account, we believe that YbZnGaO₄ is a spin glass, with frozen, short-range correlations below the freezing temperature T_f (refs 31–33). Such a phase can be identified from the a.c. susceptibility. Thus, we perform the measurements with temperatures spanning about 3 decades, ranging from 0.05 to 4 K. The results are shown in Fig. 4(c). At a measuring frequency of 100 Hz, the real part of the susceptibility χ' shows a broad peak at $T_f \approx 0.093(6)$ K. The peak height decreases, and the peak temperature increases, with increasing driving frequency f . The frequency dependence of T_f is shown in Fig. 4(b). As a quantitative measure, $\Delta P = \frac{\Delta T_f}{T_f \Delta \lg(f)}$ is 0.053(2) with f changing from 100 to 10000 Hz. This value is close to those observed in other insulating spin glasses[31–35]. The strong frequency dependence evidences a broad distribution of the spin relaxation times around T_f , typical for a spin glass[31–33].

We also measure the a.c. susceptibility for YbMgGaO₄. As shown in Fig. 4(d), the behaviors are similar to those of YbZnGaO₄, albeit with a slightly higher T_f of 0.099(6) K at 100 Hz. For YbMgGaO₄, the peak shift from 100 to 10000 Hz, ΔP is

0.068(4) [Fig. 4(b)], larger than that for YbZnGaO₄. In the insets of Fig. 4(c) and (d), we plot χ' in the whole temperature range measured. At high temperatures, it follows the Curie-Weiss law. Below ~ 2 K, it rises more rapidly with cooling. Remarkably, this temperature coincides with T^+ , below which the magnetic specific heat decreases.

The spin-glass phase identified from the a.c. susceptibility is a natural ground state for YbZnGaO₄: i) Disorder and frustration, the two ingredients for a spin glass[31–33, 36], are present and strong in such materials[4, 5, 8, 11, 16–22, 37, 38]; ii) A spin glass maintains short-range spin-spin correlations[31–33], consistent with the absence of a long-range magnetic order; iii) The observed INS spectra can be nicely reproduced by bringing disorder into an ordered state; iv) Macroscopically, a spin glass is disordered, and thus the magnons do not conduct heat due to the short mean-free path. This explains the lack of contribution to the thermal conductivity from the gapless magnetic excitations; v) Finally, we estimate the fraction of frozen moment to be 13(3)% from our INS results shown in Fig. S4, close to the 16(3)% in YbMgGaO₄ (ref. 5), but smaller than the 33% expected from theory[39]. We consider it to be a consequence of the strong frustration in this compound.

We also note that some findings in YbZnGaO₄ suggest deviations from a generic spin glass. For instance, T^+ is about 20 times of T_f , much larger than that expected for a typical spin glass[31–33]. We believe that this indicates the existence of strong frustration. In addition to the geometrical frustration inherent to the triangular structure[1, 2], the spin-space anisotropy induced by the spin-orbit coupling of the Yb³⁺ ions, recognized in our anisotropic spin model and in refs 4, 5, 11, 18–22, should further reduce T_f . Moreover, the strong disorder[5, 8, 11, 16, 17] is expected to result in a reduced T_f .

In summary, we have successfully grown high-quality single crystals for YbZnGaO₄, and our comprehensive measurements on these crystals provide concrete evidence that it is a spin glass. We suggest that this phase is driven by disorder and frustration. Our work reveals the very similar characteristics between QSL and spin-glass phases, including the broad “continuum” of magnetic excitations and is a cautionary tale about labelling such materials without a full study of the spin system.

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Supplementary Materials

Detailed methodology

Polycrystalline samples of YbZnGaO_4 (LuZnGaO_4) were synthesized by reactions of stoichiometric amounts of Yb_2O_3 (99.99%) (Lu_2O_3 , 99.99%), ZnO (99.99%), and Ga_2O_3 (99.999%) powders at 1350 °C. High-quality single crystals of YbZnGaO_4 and YbMgGaO_4 were grown by the floating-zone technique in an optical image furnace (MF-2400, Cyberstar Corp.) under a pressure of 10 bar. The chamber was filled with mixed gas of Ar and O_2 in the ratio of 4 : 1. During the growth, the seed and feed rods spun in opposite directions with a speed of 20 r/min, and traveled downwards with a speed of 1.2 mm/h.

X-ray diffraction data were collected in an x-ray diffractometer (X'TRA, ARL) using the Cu-K_α edge with a wave length of 1.54 Å. Rietveld refinements on the data were run in the Fullprof. suite. Energy dispersive x-ray spectra were collected in an energy analyser (EX-250, Horiba) equipped in a scanning tunneling microscope (S-3400N II, Hitachi). The d.c. magnetic susceptibility was measured in a physical property measurement system that cooled to 1.8 K (PPMS-9T, Quantum Design). Magnetizations with fields up to 14 T were collected in a PPMS-14T. Specific heat was measured in a PPMS-9T equipped with a dilution refrigerator. Since YbZnGaO_4 has large specific heat and long thermal relaxation time, to make sure that the sample really reached the ultralow temperature, we used as small amount of sample as possible (0.4 mg). We set a sequence that made 6 initial tests to obtain parameters for each data point, and then picked up the most appropriate ones for the measurement. Below 0.2 K, each data point took about 1 hour. Thermal conductivity was measured in a dilution refrigerator (KelvinoxMX400, Oxford Instruments). For the measurement, a standard four-wire steady-state method was employed. We made contacts on a thin-bar sample with dimensions of $2.80 \times 0.80 \times 0.14 \text{ mm}^3$ using silver epoxy. To ensure better contact, we annealed the sample at 400 °C for 30 minutes. Two RuO_2 chip thermometers were calibrated *in-situ* against a reference RuO_2 thermometer. The a.c. susceptibility measurements were conducted in a Quantum Design PPMS DynaCool with a dilution refrigerator insert (ACDR measurement option) allowing for continuous operation from 4 K down to 0.05 K (ref. 1). The sample was attached to the sapphire holder with GE-varnish. For better thermal anchoring, we used as possibly little amount of GE-varnish as to be enough to maintain a good contact with the holder. To reach ultralow temperatures, we cut the samples to as thin as possible. For the measurements, we used a sample of 29.3 mg for YbZnGaO_4 and 19.2 mg for YbMgGaO_4 .

INS experiments on the single crystals were carried out on PANDA, a cold triple-axis spectrometer located at MLZ at Garching, Germany[2]. In the measurements, we used 11 single crystals weighed 1.2 g in to-

tal, coaligned with a neutron Laue diffractometer NLaue also located at MLZ. As shown in Fig. S3(a), the full width at half maximum of the rocking scan through the (1, -1, 0) peak is 0.98° , indicating a good coalignment of the crystals. The crystals were mounted onto an aluminium sample holder with the c axis oriented along the vertical direction, such that $(H, K, 0)$ was the scattering plane. A closed-cycle refrigerator (CCR) equipped with a ^3He insert was used for cooling down the sample with the holder. We used a fixed-final-energy (E_f) mode with $E_f = 5.1 \text{ meV}$. To reduce higher-order neutrons, one 18 cm long Be filter cooled to 40 K was placed after the sample. A double-focusing mode without additional collimators was used for both the monochromator and analyser. Under such conditions, the energy resolution was about 0.1 meV (half width at half maximum, HWHM). Experiments on the 14-g powder sample were carried out on SIKA, a cold triple-axis spectrometer located at ANSTO at Lucas Heights, Australia. The powders were loaded into an aluminium can, which was then mounted onto a CCR. A Be filter was placed after the sample to reduce contaminations from higher-order neutrons. The beam collimations were $40'-40'$ -sample- $60'-40'$. The beam was vertically focused on the monochromator side. A fixed- E_f mode with $E_f = 5 \text{ meV}$ was used in the measurements. The resulting energy resolution was about 0.08 meV (HWHM). The wave vector $\mathbf{Q}$ was expressed as (H, K, L) reciprocal lattice unit (r.l.u.) of $(a^*, b^*, c^*) = (4\pi/\sqrt{3}a, 4\pi/\sqrt{3}b, 2\pi/c)$, with $a = 3.414(2) \text{ Å}$ and $c = 25.140(2) \text{ Å}$.

Structure and element analyses

We have obtained single-crystal samples of YbZnGaO_4 using the floating-zone technique. Some of the crystals are shown in Fig. S1(a). We ground some single crystals into fine powders for the x-ray diffraction (XRD) measurements. The XRD results on the powders measured at room temperature are presented in Fig. S1(b). From the data, we do not observe any noticeable intensities from a second phase. The data can be well refined with an YbFe_2O_4 -type crystal structure (space group $R\bar{3}m$, No. 166). The least-square fit results in $\chi^2 \approx 1.07$, indicating an excellent agreement of the fit with the data. The lattice constants resulting from the fit are $a = 3.414(2) \text{ Å}$, $c = 25.140(2) \text{ Å}$. Other results from the refinements are tabulated in Table S1.

TABLE S1. Rietveld refinement results at room temperature

Atom	Wyckoff position	(0, 0, z)	Occ
Yb	3a	0	1
Zn	6c	0.2142(3)	1
Ga	6c	0.2142(3)	1
O(1)	6c	0.2894(2)	1
O(2)	6c	0.1284(6)	1

Energy dispersive x-ray spectroscopy results on the sin-

gle crystals are presented in Fig. S1(c). By analyzing the spectra, we obtain the chemical contents as shown in Table S2. These results show that the chemical compositions are close to the stoichiometric values.

TABLE S2. Element contents analysed from the energy dispersive x-ray spectra

Element	Mass percent	Atomic percent
Zn	20.2(2)	32.6(3)
Ga	20.9(3)	31.5(2)
Yb	58.9(3)	35.9(3)
Total	100.00	100.00

Additional d.c. magnetization and specific heat results

We have measured both the single-crystal and polycrystalline samples with magnetic fields up to 14 T at $T = 2$ K, and the results are presented in Fig. S2(a). Above approximately 11 T, the magnetization depends linearly on the magnetic field as $M = \chi^{\text{VV}} \mu_0 H + g \mu_B S$. The slope χ^{VV} is the temperature-independent paramagnetic Van Vleck susceptibility[3, 4]. In our case, the effective spin S is 1/2. By fitting the data above 11 T, for the single crystal, we obtain $\chi_{\perp}^{\text{VV}} = 0.0019(3)$ cm³ per mol Yb³⁺ for fields perpendicular to the c axis, and $\chi_{\parallel}^{\text{VV}} = 0.0561(3)$ cm³ per mol Yb³⁺ for fields parallel; for the powder sample, $\chi_{\text{p}}^{\text{VV}} = 0.0519(3)$ cm³ per mol Yb³⁺. The susceptibility data presented in Fig. 1(c) have been corrected using these values. From the fits, we also obtain the saturated moments $g_{\perp} \mu_B/2$, $g_{\parallel} \mu_B/2$, and $g_{\text{p}} \mu_B/2$, and thus Landé g factors of $g_{\perp} = 3.17(4)$, $g_{\parallel} = 3.82(2)$, and $g_{\text{p}} = 3.58(3)$. These values are listed in Table I.

The specific heat for the polycrystalline sample of YbZnGaO₄ measured at several magnetic field strengths are presented in Fig. S2(b). Measurements on a single crystal have also been performed, and the results at zero field overlap with those of the polycrystalline sample [Fig. S2(b)]. Specific heat of the reference sample, a nonmagnetic LuZnGaO₄ is also shown. The specific heat for LuZnGaO₄ can be well fitted with the Debye model as $C \sim T^3$. The data for LuZnGaO₄ is used as the lattice contribution to the total specific heat for YbZnGaO₄. By subtracting the phonon contribution, we obtain the magnetic specific heat for YbZnGaO₄ as shown in Fig. 1(d).

More neutron scattering results

The high quality of the single crystals is demonstrated in Fig. S3(a), in which we plot the rocking scan through the (1, -1, 0) nuclear Bragg peak for the 11 coaligned single crystals. The peak is reasonably sharp with a full width at half maximum of 0.98°. For the scattering at $E = 0.3$ meV [Fig. 2(a)], the strong intensities at (-0.5, 0.5, 0) are indeed contaminated by the elastic scattering with second-order neutrons of the (-1, 1, 0) nuclear

Bragg peak. This is testified in Fig. S3(b), where we compare the intensities for two nuclear Bragg peaks (1, -1, 0), and (0, 1, 0). For (1, -1, 0), symmetrically equivalent to (-1, 1, 0), intensities are strong, so as for (-0.5, 0.5, 0). On the other hand, for (0, 1, 0), intensities are weak, and so as for (0, 0.5, 0). Furthermore, the additional feature at (-0.5, 0.5, 0) is suppressed with increasing energies, and disappears at $E = 0.6$ meV, while magnetic excitations remain significant. In fact, (1, -1, 0) and (0, 1, 0) are not allowed, but due to the large c -axis lattice constant, there is a small out-of-plane component of $L = 1$ in these peaks, making these peaks detectable.

In Fig. S4(a), we plot energy scans at three representative $\mathbf{Q}$ points, M₁, K, and Γ_1 . It clearly shows that the spectral weights at M₁ and K are the same within experimental error, but significantly higher than those at Γ_1 . The intensities decrease from the lowest energy measured, and remain finite up to ~ 1.4 meV, similar to YbMgGaO₄ as in refs [5, 6]. We have also performed INS measurements on the powder sample of YbZnGaO₄, and the results are shown in Fig. S4(b). On the energy-loss side ($E > 0$), intensities increase with decreasing temperatures; on the energy-gain ($E < 0$) side, the temperature dependence is opposite. These results indicate that the intensities are resulting from intrinsic magnetic excitations. We observe magnetic excitations down to $E = 0.2$ meV, below which the signals are contaminated by the elastic scattering. We estimate the spectral weight in the elastic channel (≤ 0.1 meV) to be 13(3)% of the total spectral weight. The value is close to the 16(3)% found in YbMgGaO₄ (ref. [6]), but smaller than the 33% expected from theory[7]. We fit the data in Fig. S4(b) in an energy range of 0.2 to 2 meV, and the resulting integrated intensities are plotted against the temperature in Fig. S4(c). The intensities grow rapidly below ~ 10 K, indicating that the short-range correlations between spins occur at a much higher temperature than the freezing temperature.

Linear spin-wave calculations

In the calculations, we use an anisotropic spin model with nearest-neighbor (NN) and next-nearest-neighbor exchange interactions. Within this model, we introduce disorder into the stripe-order phase by the Holstein-

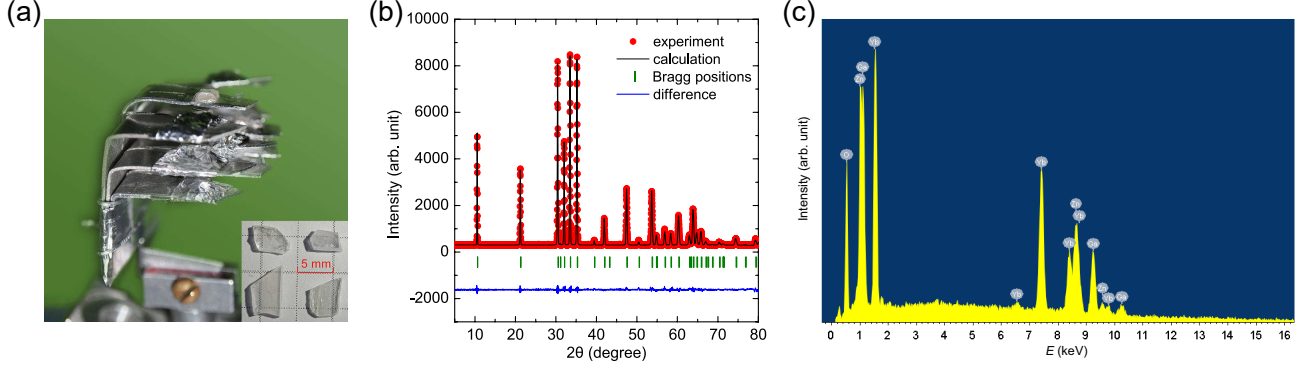


FIG. S1. (a) A photograph of 11 coaligned single crystals weighing 1.2 g in total, mounted onto an aluminium sample holder. The inset shows 4 single crystals before the coalignment. (b) Powder x-ray diffraction data and the calculations. (c) Energy dispersive x-ray spectra. Elements are marked on top of the corresponding peaks.

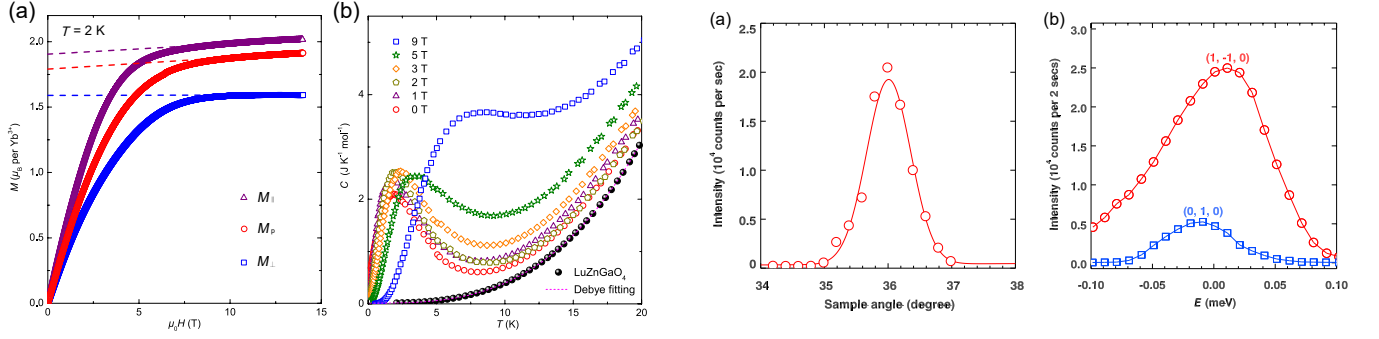


FIG. S2. (a) Magnetic-field dependence of the magnetization for the single-crystal (for fields parallel and perpendicular to the c axis, $M_{||}$ and $M_{\perp}$), and polycrystalline sample (M_p), measured at $T = 2$ K. Dashed lines are linear fits to the data for $\mu_0 H \geq 11$ T. (b) Total specific heat of a YbZnGaO_4 polycrystalline sample measured at various magnetic fields, and a single crystal measured at zero field. Specific heat of a reference sample, nonmagnetic LuZnGaO_4 , is also shown. The specific heat for LuZnGaO_4 can be nicely fitted by the Debye model as $C \sim T^3$, indicated by the dashed line.

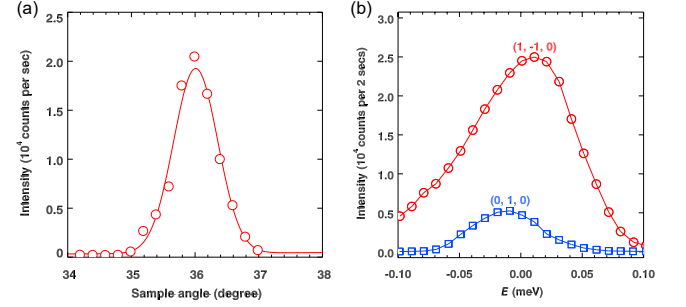


FIG. S3. (a) Rocking scan through the $(1, -1, 0)$ nuclear Bragg peak of the 11 coaligned single crystals. The line through data is a fit with a Gaussian function. (b) Quasielastic scans through two nuclear Bragg peaks $(1, -1, 0)$, and $(0, 1, 0)$. Lines through data are guides to the eye.

Primakoff transformation[8], and obtain

$$\begin{aligned}
 H = & \sum_i u(1 + \Delta_i)b_i^\dagger b_i \\
 & + \sum_{\langle ij \rangle_{||}} [t_1(1 + \Delta_{ij})b_i^\dagger b_j + p_1(1 + \Delta_{ij})b_i^\dagger b_j^\dagger + h.c.] \\
 & + \sum_{\langle ij \rangle_{\#}} [t_2(1 + \Delta_{ij})b_i^\dagger b_j + p_2(1 + \Delta_{ij})b_i^\dagger b_j^\dagger + h.c.] \\
 & + \sum_{\langle\langle ij \rangle\rangle_{\uparrow\uparrow}} \frac{J_2}{2} [(1 + \Delta_{ij})b_i^\dagger b_j + \Delta_{ij}b_i^\dagger b_j^\dagger + h.c.] \\
 & + \sum_{\langle\langle ij \rangle\rangle_{\uparrow\downarrow}} \frac{J_2}{2} [(1 + \Delta_{ij})b_i^\dagger b_j^\dagger + \Delta_{ij}b_i^\dagger b_j + h.c.]. \quad (S1)
 \end{aligned}$$

Here, $b_i^\dagger$ and b_i are Holstein-Primakoff bosonic operators, Δ_i and Δ_{ij} represent the site and bond disorder,

$\langle ij \rangle_{||}$ and $\langle ij \rangle_{\#}$ indicate NN pairs parallel and unparallel to the stripe order, and $\langle\langle ij \rangle\rangle_{\uparrow\downarrow}$ and $\langle\langle ij \rangle\rangle_{\uparrow\uparrow}$ indicate NNN pairs between stripes with opposite and same spin directions. The NN hopping and pairing parameters are obtained from the spin exchange interaction by the

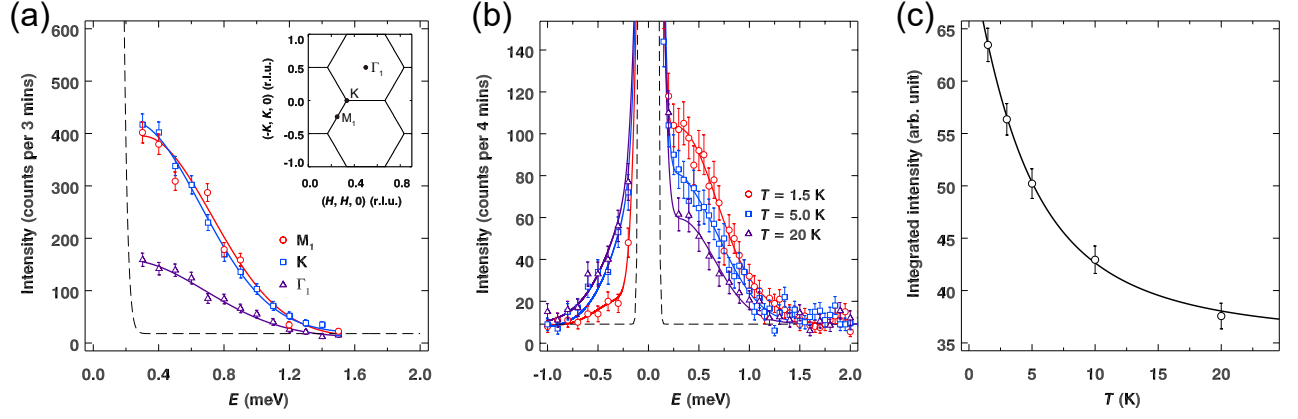


FIG. S4. (a) Energy scans at M₁, K, and Γ₁ points for the single crystals. Lines through data are fits with Lorentzian functions. The dashed line indicates the instrumental resolution. The inset indicates the $\mathbf{Q}$ positions. (b) Energy scans at $\mathbf{Q} = 1.067 \text{ \AA}^{-1}$ (M points of the first Brillouin zone) at three temperatures for the powder sample. Lines through data are fits with a Lorentzian function convoluted with a Gaussian function, latter of which represents the instrumental resolution as indicated by the dashed line. (c), Integrated intensities plotted as a function of temperature, obtained by fitting the scans in (b) in an energy range of 0.2 to 2 meV at different temperatures. The line through data is a guide to the eye.

Holstein-Primakoff transformation as,

$$\begin{aligned}
 u &= (J_{zz} + J_2)\cos^2\theta + J_{z\pm}\sin 2\theta \\
 &\quad + (J_2 + 2J_{\pm} - 2J_{\pm\pm})\sin^2\theta, \\
 t_1 &= \frac{1}{4}[2(J_{\pm} + J_{\pm\pm}) + 2(J_{\pm} - J_{\pm\pm})\cos^2\theta \\
 &\quad - J_{z\pm}\sin 2\theta + J_{zz}\sin^2\theta], \\
 t_2 &= \frac{1}{4}[-2J_{\pm} + J_{\pm\pm} + (2J_{\pm} + J_{\pm\pm})\cos^2\theta + J_{zz}\sin^2\theta \\
 &\quad + \frac{1}{2}J_{z\pm}\sin 2\theta + i\sqrt{3}(J_{z\pm}\sin\theta - 2J_{\pm\pm}\cos\theta)], \\
 p_1 &= \frac{1}{4}[-2(J_{\pm} + J_{\pm\pm}) + 2(J_{\pm} - J_{\pm\pm})\cos^2\theta \\
 &\quad - J_{z\pm}\sin 2\theta + J_{zz}\sin^2\theta], \\
 p_2 &= \frac{1}{4}[2J_{\pm} - J_{\pm\pm} + (2J_{\pm} + J_{\pm\pm})\cos^2\theta \\
 &\quad + \frac{1}{2}J_{z\pm}\sin 2\theta + J_{zz}\sin^2\theta],
 \end{aligned}$$

where $\theta = \frac{\pi}{4} + \frac{1}{2}\tan^{-1}\left(\frac{J_{\pm} + 2J_{\pm\pm} - J_{zz}/2}{2J_{z\pm}}\right)$, which indicates the direction of the local magnetic order due to the anisotropic interactions. Employing the linear spin-wave theory[9], we calculate the spin-spin correlation function,

$$I(\mathbf{Q}, E) = \frac{1}{N} \sum_{ij} e^{i\mathbf{Q}(\mathbf{r}_i - \mathbf{r}_j)} \int_{-\infty}^{\infty} \langle \mathbf{S}_i \cdot \mathbf{S}_j(t) \rangle e^{-iEt} dt, \quad (\text{S2})$$

where $\mathbf{S}_i$, which should be expressed as bosons operators $b^\dagger$ and b in the calculation, is the effective spin at site i with the coordinate $\mathbf{r}_i$, and N is the total number of sites. The calculations are performed over 32×32 triangular-

lattice sites. An example of the calculation results is shown in Fig. 3, in which we use a set of parameters with random Δ_i and Δ_{ij} satisfying a Gaussian distribution with the variance of 0.2, $J_{zz} = 1.6J_{\pm}$, $J_{\pm\pm} = J_{z\pm} = 0.06J_{\pm}$, and $J_2 = 0.15J_{\pm}$.

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